

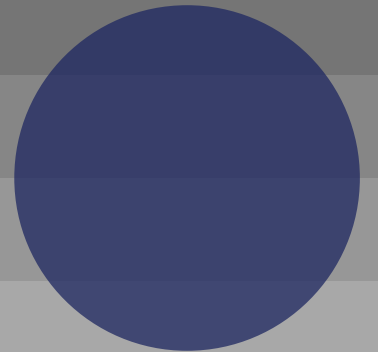
IMAGE PROCESSING AND SEGMENTATION

Thresholding Approach to Improve Segmentation Accuracy in Grayscale Images



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Introduction

Understanding grayscale images and why segmentation matters



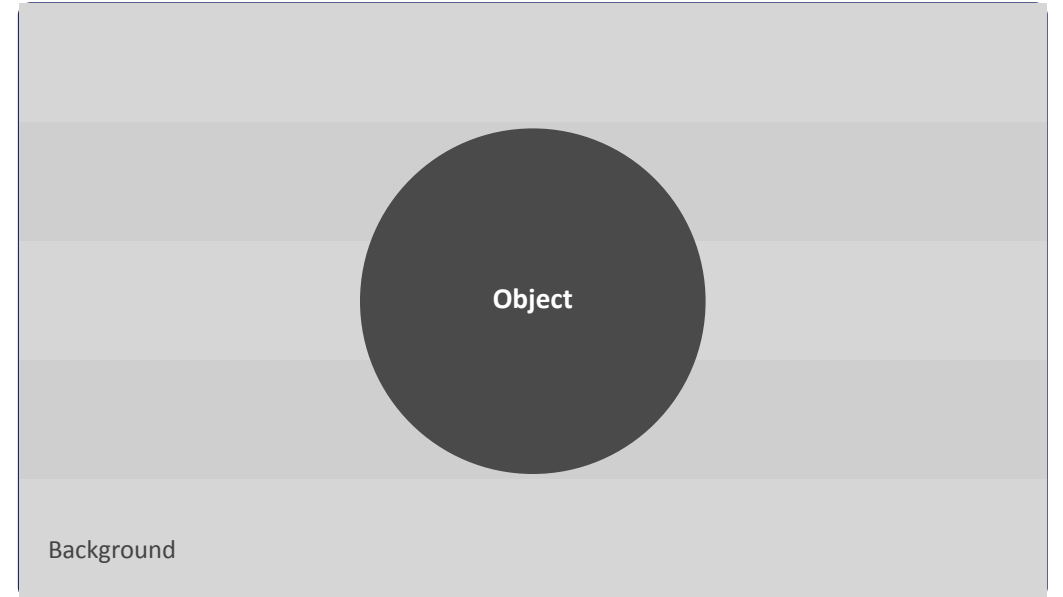
Grayscale Image

A single-channel image where each pixel holds an intensity value from 0 (black) to 255 (white) — no color information.



Image Segmentation

The process of dividing an image into meaningful regions, separating the object of interest from the background.



Example: Grayscale Image (Object vs. Background)

Why Segmentation Matters



Medical
Imaging



Autonomous
Vehicles



Industrial
Inspection



Satellite
Imaging



Computer
Vision

Problem Statement

Poor intensity separation between object and background limits segmentation accuracy



Low Contrast

Object and background intensities lie too close together.



Noise

Random pixel variations obscure true intensity values.



Uneven Illumination

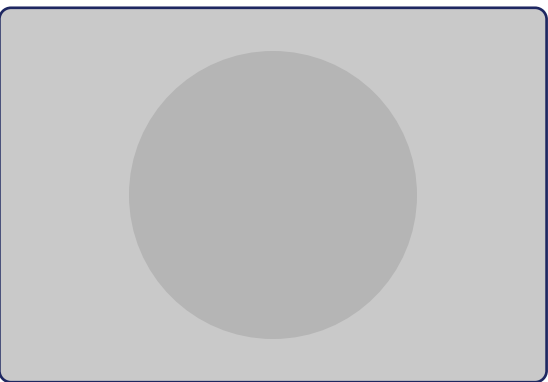
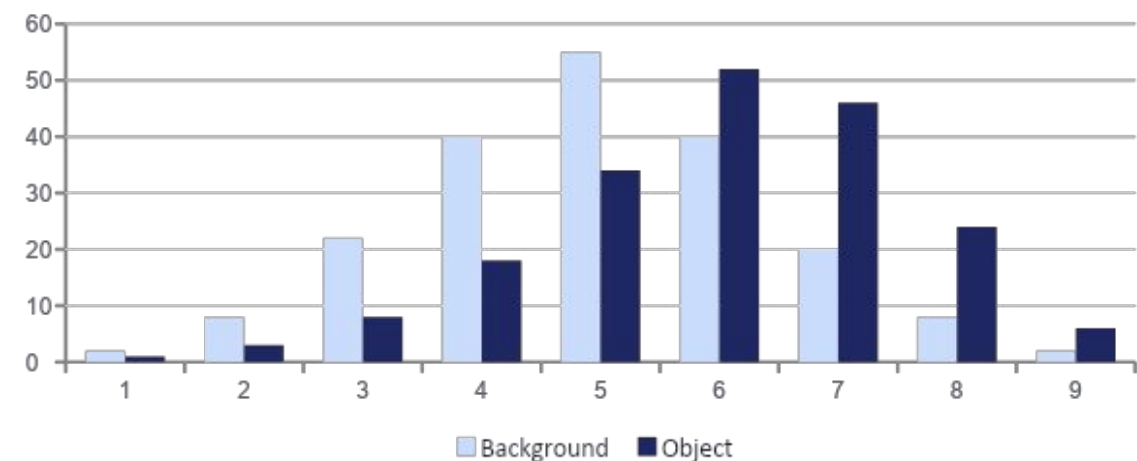
Lighting varies across the image, shifting intensities.



Overlapping Intensity Values

Object and background histograms overlap significantly.

Overlapping Intensity Histograms



Blurred boundary —
object barely visible

**Result: Inaccurate
Segmentation Boundary**

Thresholding Approach

Converting a grayscale image into a binary image using an intensity threshold (T)



Global Thresholding

One fixed threshold value applied to every pixel in the image.



Adaptive Thresholding

Threshold varies across small regions to handle uneven lighting.



Otsu's Thresholding

Automatically computes the optimal threshold from the histogram.

Comparison of Thresholding Methods

Method	Threshold Selection	Best Suited For	Accuracy
Global	Manual / fixed value	Uniform lighting	Moderate
Adaptive	Local, region-based	Uneven illumination	Good
Otsu's	Automatic, histogram-based	Poor object–background separation	High

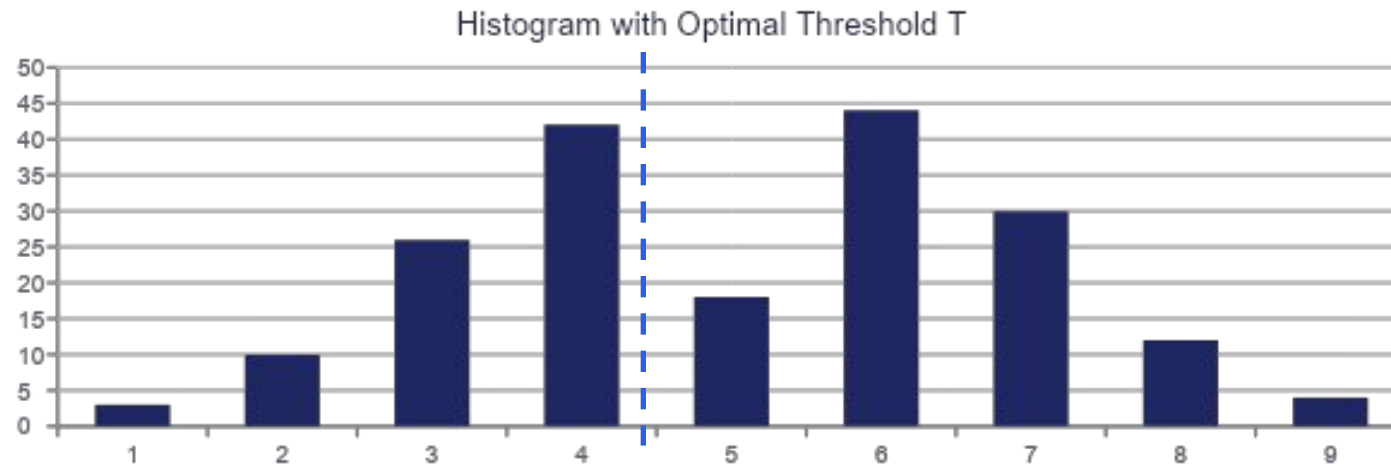
Why Otsu's Thresholding: It selects the threshold automatically and objectively, making it the most reliable choice when object and background intensities are poorly separated.

Working of Otsu's Thresholding

A step-by-step automatic process for finding the optimal threshold



Otsu's method finds the threshold T that maximizes between-class variance, separating the histogram into two distinct classes.



Advantages and Results

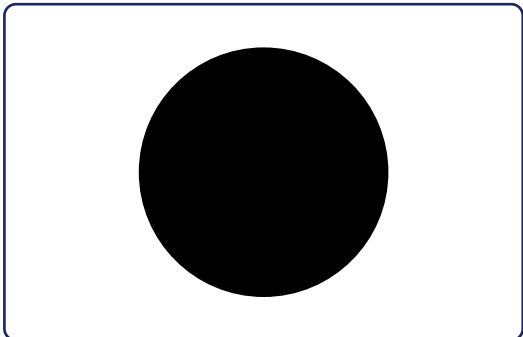
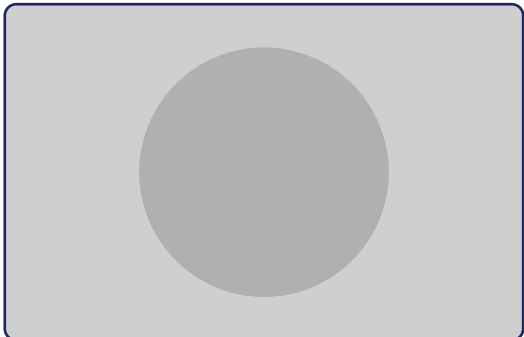
Advantages

- ✓ Automatic threshold selection
- ✓ Better segmentation accuracy
- ✓ Easy to implement
- ✓ Fast computation
- ✓ Reduces manual effort
- ✓ Widely used in image processing

Applications

- Medical image analysis
- Face recognition
- Optical character recognition (OCR)
- Object detection
- Industrial quality inspection
- Satellite image analysis

Before and After Segmentation



Otsu's Thresholding produces a sharp, accurate object boundary from a low-contrast grayscale input.

Conclusion



Thresholding is a simple yet powerful technique that converts a grayscale image into a clear, binary segmentation of object and background.



Otsu's Thresholding improves segmentation accuracy by automatically selecting the optimal threshold from the image histogram — no manual tuning needed.



Selecting the correct threshold significantly improves object detection, boundary accuracy, and overall image analysis quality.

Accurate thresholding = Accurate segmentation = Reliable image analysis

References

IEEE Style



- [1] R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed. New York, NY, USA: Pearson, 2018.
- [2] N. Otsu, "A threshold selection method from gray-level histograms," IEEE Trans. Syst., Man, Cybern., vol. 9, no. 1, pp. 62–66, Jan. 1979.
- [3] OpenCV Development Team, "Image Thresholding," OpenCV Documentation. [Online]. Available: <https://docs.opencv.org>
- [4] The MathWorks Inc., "Image Processing Toolbox," MATLAB Documentation. [Online]. Available: <https://www.mathworks.com>
- [5] IEEE Xplore Digital Library. [Online]. Available: <https://ieeexplore.ieee.org>